

CURRENT PROTOTYPE ARCHITECTURE



PROOFCOACH AI

Build it. Prove it. Defend it.

A Simple Guide to How the System Works — Explaining how resume parsing, role evidence, adaptive interview defense, and targeted learning connect together.

Document Type	Report Date	Architecture
System & Product Guide	September 2026	Local-First / Deterministic Slice

What Is ProofCoach AI?

ProofCoach AI is a privacy-first career preparation system designed to help students, freshers, and technical candidates bridge the gap between *writing* an impressive resume and *defending* their skills in real technical interviews.

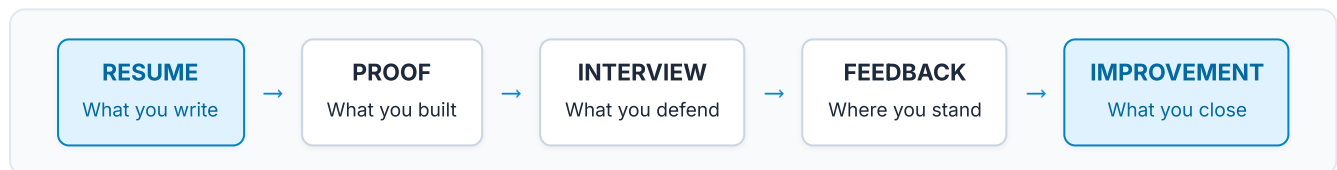
The Core Problem

In the era of generative AI, anyone can generate a resume full of impressive-sounding keywords and metrics (e.g., "*Reduced API latency by 35%*"). However, candidates frequently struggle when interviewers challenge those exact claims:

The Preparation Gap: Candidates polish resume wording, but have never practiced explaining their baseline measurements, architectural trade-offs, or personal contributions under pressure.

The ProofCoach Solution

ProofCoach connects the candidate's journey into an unbroken, evidence-backed chain:



Who Is It For?

Students

Translate university coursework and academic projects into defensible, commercial-grade evidence.

Freshers & New Grads

Overcome limited work history by building verified project artifacts that bypass automated parsing barriers.

Early-Career Engineers

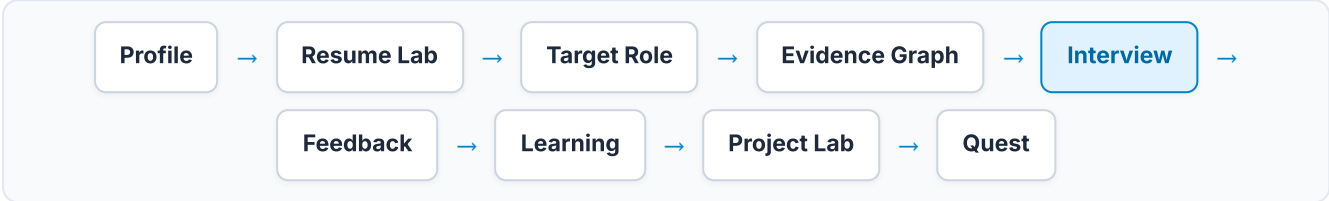
Practice articulating personal ownership, system trade-offs, and optimization benchmarks.

Career Switchers

Identify critical role gaps and map existing transferable skills to new target technical roles.

The Complete Product in One View

ProofCoach AI organizes technical preparation into an integrated workflow where every tool directly feeds the next:



 **Profile & Command Centre**
Stores career goals and daily study limits privately, displaying six clear diagnostic metrics.

 **Resume Lab**
Tests resume machine-readability (Parser Robustness) and automatically detects measurable claims.

 **Target Role**
Extracts job requirements from job descriptions and highlights your skill coverage and critical gaps.

 **Career Evidence Graph**
Visually links skills and projects to claims and job specs, exposing missing evidence chains.

 **Adaptive Interview**
Interrogates you directly on your resume claims, adapting difficulty based on your answers.

 **Evidence Feedback**
Scores your answers across 6 dimensions and uses **Evidence Lock** to block fabricated rewrites.

 **Personalized Learning**
Turns identified skill gaps into concrete study tasks linked directly to free, official documentation.

 **Project Lab**
Guides you to build real software artifacts rather than passively watching video tutorials.

 **Video Study Planner**
Breaks long tutorials into daily study chunks with timestamped notes and a 75/25 study-practice split.

 **Career Quest**
Gamifies verified progress across three levels (Build, Prove, Perform) with action-linked XP.

Supporting Governance: *Focus Shield* provides a 2.5-hour wellbeing cap with session autosave; *Settings* offers one-click complete local data deletion and instant dark/light theme switching.

What Happens Inside?

ProofCoach AI connects your documents, goals, and interview performance through an automated internal processing chain. Here is what happens behind the scenes:

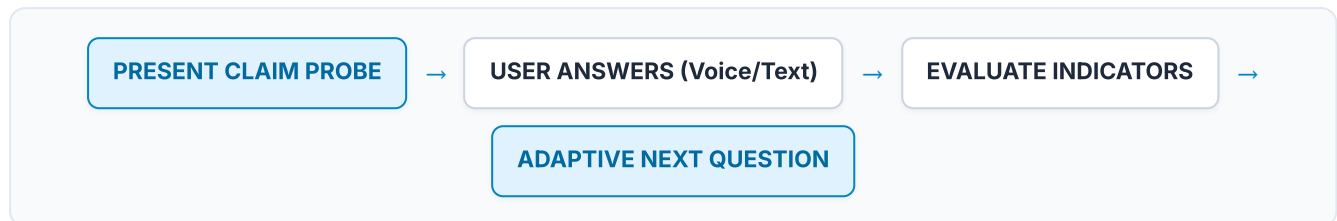


Step-by-Step Flow Explanation

- 1 Resume Upload & In-Memory Extraction:** The user selects a PDF or DOCX file. The system processes it in RAM—files are never permanently stored on a cloud server.
- 2 Parser Robustness Diagnostics:** Software inspects reading order, contact details, dates, and warns if two-column layouts or tables may confuse automated parsers.
- 3 Claim Detection:** The engine scans for quantified statements (e.g., *"Reduced API latency by 35%"*) that require technical defense.
- 4 Role Matching & Gap Identification:** Candidate skills are compared against role requirements (e.g., Backend Developer). Covered skills and missing gaps (e.g., Docker, Testing) are highlighted.
- 5 Targeted Adaptive Interview:** The interviewer challenges the candidate on their specific claim: *"What was the baseline? How did you measure it? What did you personally change?"*
- 6 Grounded Evaluation & Feedback:** The answer is evaluated for baselines, personal ownership, and trade-offs. Weak areas become prioritized learning and project tasks.

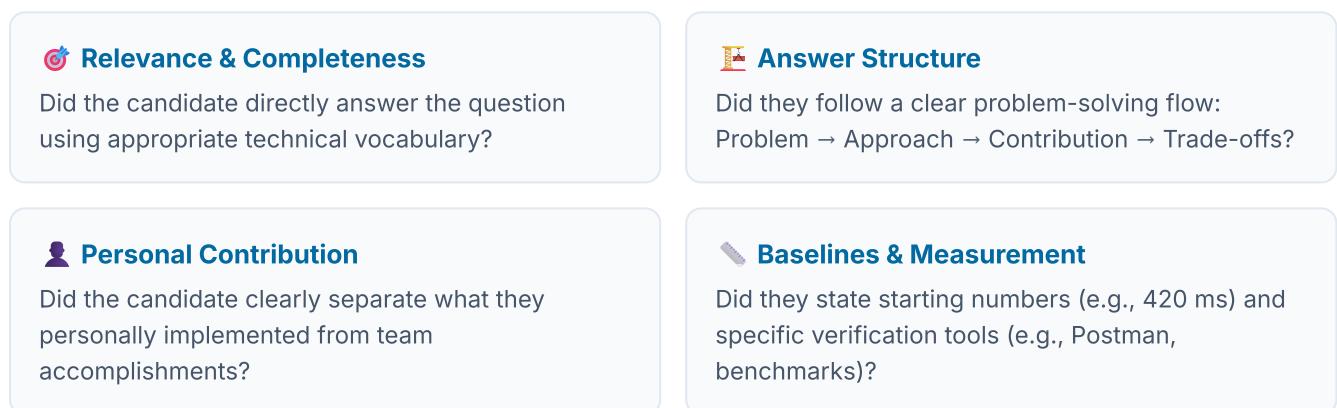
How the Interview Works

Instead of asking generic trivia questions, the **Live Adaptive Interview** interrogates the candidate directly on the achievements they claimed on their resume.

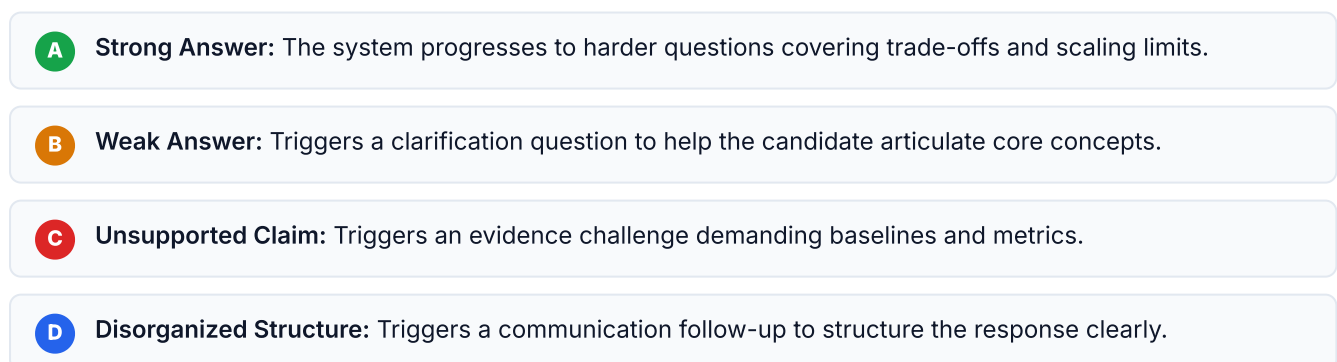


What the System Evaluates (Observable Indicators)

ProofCoach evaluates observable communication and technical structure—never fake emotion analysis:



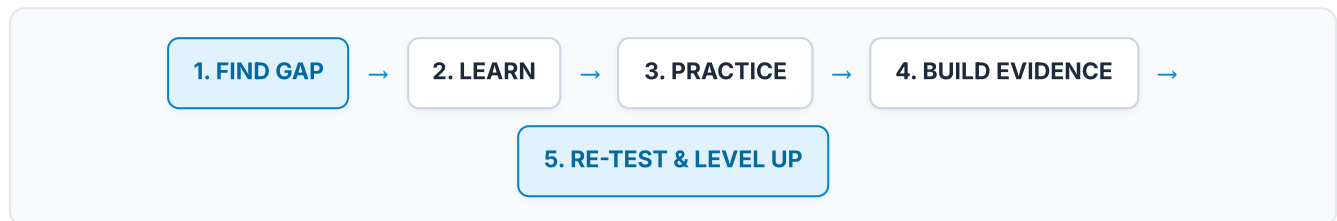
The 4-Way Adaptive Questioning Route



Ethical AI Boundary: The current prototype uses deterministic, rule-based evaluation to demonstrate the concept reliably without external AI API costs. Voice features use browser speech recognition. No facial or emotion analysis is used.

How the System Helps You Improve

ProofCoach AI is not just a diagnostic test—it is an ongoing growth loop. When an interview exposes a gap, the platform immediately channels that weakness into concrete action.



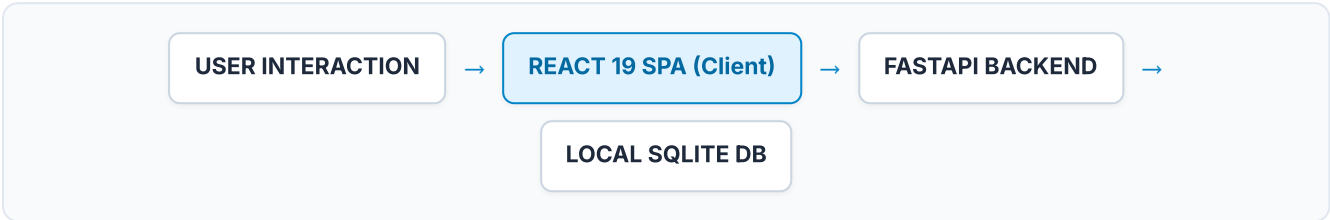
The Five Growth Tools

- 1 Personalized Learning Plan:** Replaces 40-hour video catalogs with direct, prioritized links to official documentation (e.g., Docker Get Started, pytest tutorial) for your specific missing skills.
- 2 Video Study Planner:** Embeds YouTube tutorials safely and calculates an optimal daily study schedule with an active 75% learning / 25% practice split, tracking notes by video timestamp.
- 3 Project Lab:** Guides you through a 9-stage software journey (Idea → Plan → Build → Test → Defend) with Kanban task tracking, turning study into real code evidence.
- 4 Career Quest:** Rewards verified milestones across three progressive tiers (Level 1: Build → Level 2: Prove → Level 3: Perform) with transparent, action-linked Evidence XP.
- 5 Focus Shield:** Enforces a 2.5-hour continuous study wellbeing limit with automatic session saving to prevent burnout and encourage healthy learning habits.

Evidence Lock Protection: When improving your resume wording, Evidence Lock ensures that stylistic enhancements are approved, but strictly blocks any invented numbers or unsupported technologies.

What Is Happening Technically?

ProofCoach AI runs on a clean, decoupled, local-first technical architecture designed for speed, privacy, and offline reliability.



The Five Core Technical Pillars

 **React 19 Frontend**
Single Page Application built with Vite and TypeScript. Manages local state via ProofContext and uses React Flow for graph mapping and Recharts for metrics.

 **FastAPI Python Backend**
Lightweight asynchronous service running on localhost:8000. Implements clean REST endpoints for parsing, matching, and interview evaluation.

 **In-Memory Document Parsers**
PyMuPDF and python-docx extract text and layout bounding-boxes in memory. Raw resume files are never saved permanently to disk.

 **Deterministic Rule Engines**
Pure Python services calculate Parser Robustness, match role requirements, and evaluate answer rubrics using transparent, reproducible heuristics.

Data Storage & Local-First Privacy

-  **Zero Cloud Telemetry:** No analytics SDKs (Google Analytics, Mixpanel, Sentry) are installed or imported.
-  **Local SQLite Persistence:** Profile details and interview transcripts reside exclusively on the user's computer in proofcoach.db.
-  **Zero Cloud AI Dependency:** Runs reliably offline without cloud AI API keys. An optional Ollama adapter is supported for local LLMs.

Current Status & Simple Summary

ProofCoach AI is a fully functional deterministic prototype ready for demonstration and pilot testing.

Currently Implemented & Working

✓ 13 Application Routes

Complete flow from initial onboarding through live interview, project lab, and settings.

✓ In-Memory Resume Parsing

Extracts text and checks 8 transparent robustness criteria with layout warnings.

✓ Interactive Evidence Graph

Dynamic React Flow diagram connecting skills, projects, claims, and role specs.

✓ Evidence Lock Protection

Rule-governed rewrite verification blocking fabricated metrics and tools.

Current Boundaries & Limitations

Transparent Boundaries:

- **Vercel Online Demo:** Demonstrates the frontend with deterministic fixtures; real file extraction runs with the local FastAPI backend.
- **Single-User SQLite:** Built for local-first single-user prototyping; multi-user cloud pilots will require PostgreSQL.
- **Speech Features:** Voice recognition uses the browser's Web Speech API; a reliable text fallback is always provided.

HOW PROOF COACH WORKS IN ONE SENTENCE

"ProofCoach AI connects a candidate's resume claims to evidence, tests those claims through targeted interview questions, identifies gaps, and turns those gaps into a path for improvement."